CloudNine Systems | ERD

23 tables in 3NF. Table names: singular lower_case; attributes: camelCase.

PK = primary key; FK = foreign key. No alternate-key markers are shown.

Each detail page lists every attribute and every outgoing foreign key.

Relationship lanes are independent: lines never cross.

Endpoint numbers give the number of records at that end per opposite record.

1 = exactly one; 0..1 = zero or one; 0..* = zero or more.

Nullable attributes say "optional". All other attributes are required.

Uniqueness and other business rules: analysis report, section 4 (R1-R9).

Tool: generated PDF; not a Lucidchart-authored submission.

Assignment 3 is a formatting reference; the domain remains CloudNine.

Member A

Clients and reservations

client - page 2
user_account - page 3
client_membership - page 4
project - page 5
reservation - page 12
reservation_item - page 13

Member B

Infrastructure and availability

region - page 6
availability_zone - page 7
data_center - page 8
resource_type - page 9
resource - page 10
resource_state - page 11

Member C

Deployments and services

deployment - page 14
resource_allocation - page 15
deployment_service - page 16
service - page 17
service_rate - page 18

Member D

Metering and billing

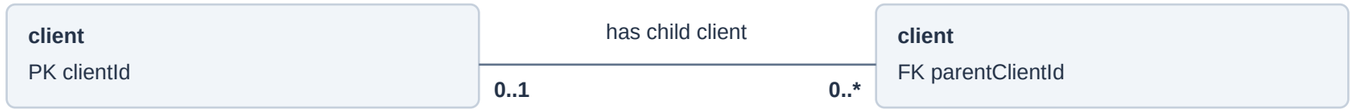
usage_record - page 19
billing_account - page 20
charge - page 21
charge_share - page 22
invoice - page 23
invoice_line - page 24

client | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**clientId**	**int**
FK	parentClientId	int (optional)
	name	text
	clientType	text
	contactEmail	text
	qualificationNote	text (optional)

Relationships (parent to child)



user_account | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**userId**	**int**
	name	text
	email	text
	accountKind	text
	staffRole	text (optional)
	status	text

Relationships (parent to child)

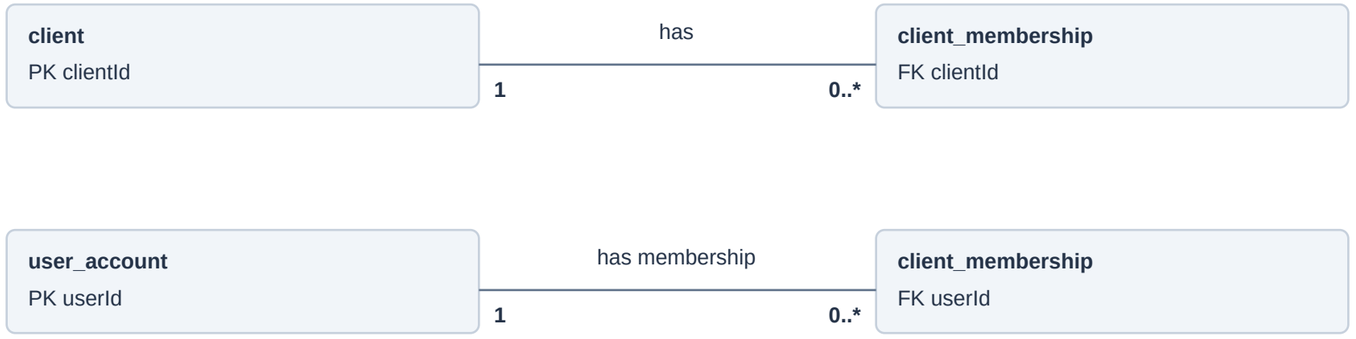
No outgoing foreign keys; incoming links appear on child pages.

client_membership | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**membershipId**	**int**
FK	clientId	int
FK	userId	int
	role	text
	validFrom	timestamp
	validTo	timestamp (optional)

Relationships (parent to child)



project | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**projectId**	**int**
FK	clientId	int
	name	text

Relationships (parent to child)

region | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**regionId**	**int**
	name	text
	regulatoryNote	text

Relationships (parent to child)

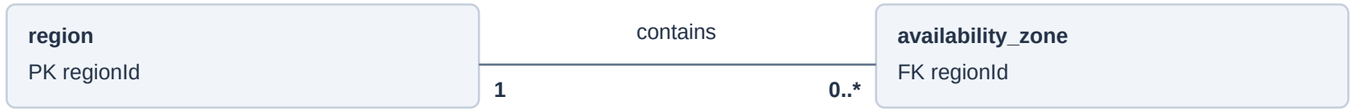
No outgoing foreign keys; incoming links appear on child pages.

availability_zone | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**zoneId**	**int**
FK	regionId	int
	name	text
	redundancyNote	text
	connectivityNote	text

Relationships (parent to child)

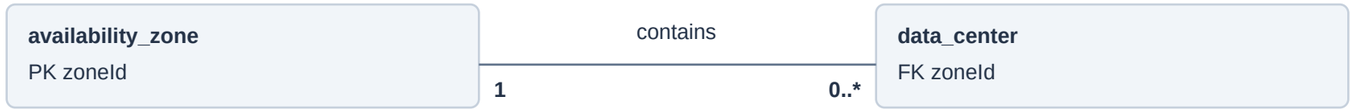


data_center | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**dataCenterId**	**int**
FK	zoneId	int
	name	text
	location	text

Relationships (parent to child)



resource_type | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**resourceTypeId**	**int**
	name	text
	intendedWorkload	text

Relationships (parent to child)

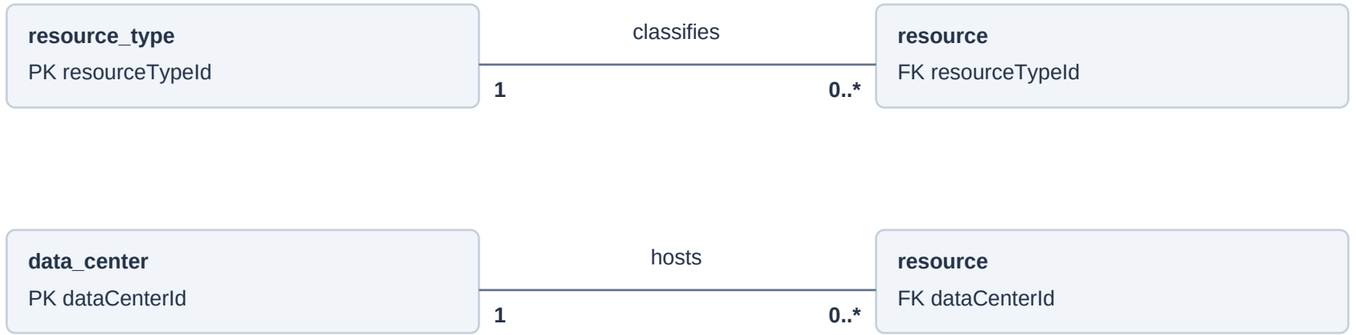
No outgoing foreign keys; incoming links appear on child pages.

resource | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**resourceId**	**int**
FK	resourceTypeId	int
FK	dataCenterId	int
	localIdentifier	text
	performanceTier	text
	vcpuCount	int
	memoryGib	decimal
	storageGib	decimal
	networkGbps	decimal

Relationships (parent to child)

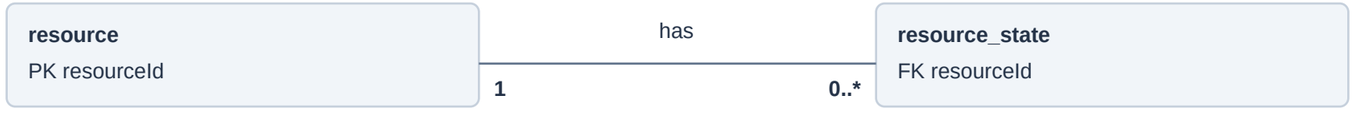


resource_state | Member B

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**stateId**	**int**
FK	resourceId	int
	status	text
	reason	text
	startsAt	timestamp
	endsAt	timestamp (optional)
FK	recordedByUserId	int (optional)

Relationships (parent to child)



reservation | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**reservationId**	**int**
FK	clientId	int
	createdAt	timestamp
	startTime	timestamp
	endTime	timestamp
	status	text
	agreementNote	text (optional)
FK	approvedByUserId	int (optional)

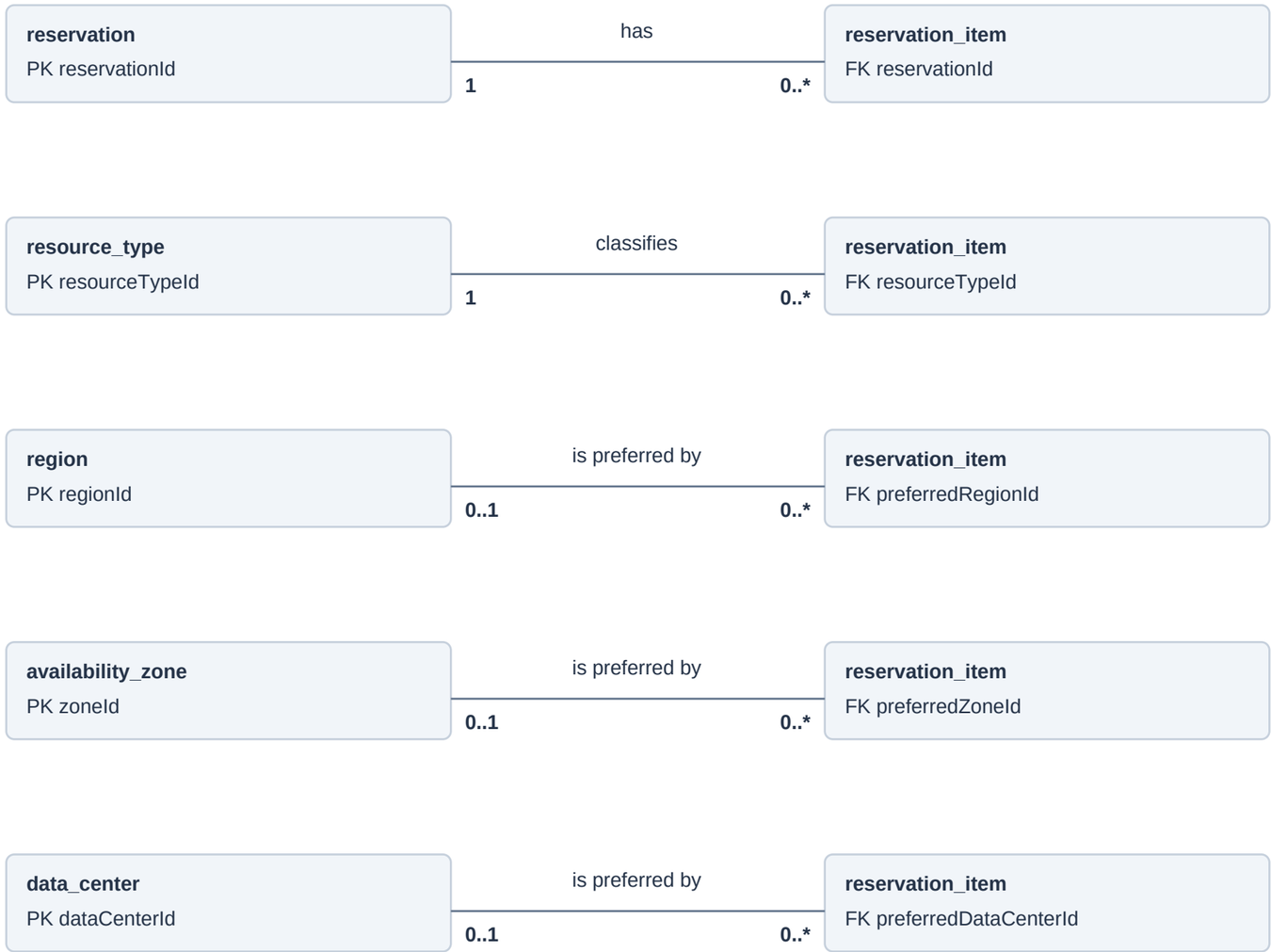
Relationships (parent to child)

reservation_item | Member A

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**reservationItemId**	**int**
FK	reservationId	int
FK	resourceTypeId	int
	quantity	int
	minVcpu	int
	minMemoryGib	decimal
	minStorageGib	decimal
	minNetworkGbps	decimal
FK	preferredRegionId	int (optional)
FK	preferredZoneId	int (optional)
FK	preferredDataCenterId	int (optional)
	locationMode	text

Relationships (parent to child)

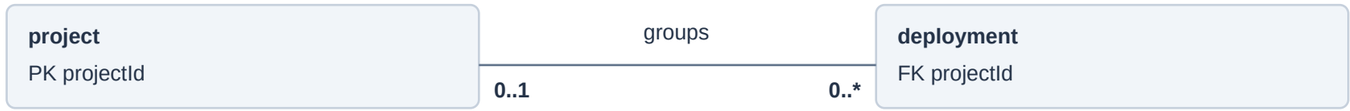


deployment | Member C

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**deploymentId**	**int**
FK	reservationId	int
FK	projectId	int (optional)
	name	text
	workloadNote	text
	startedAt	timestamp (optional)
	endedAt	timestamp (optional)
	confidentialityLevel	text
	status	text
	priority	int
	scalingMode	text

Relationships (parent to child)

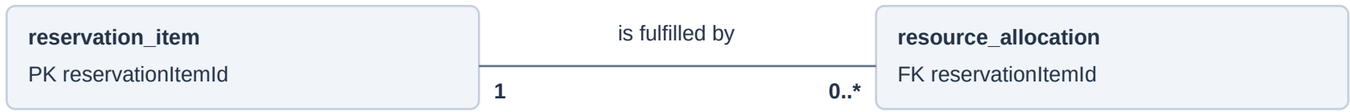


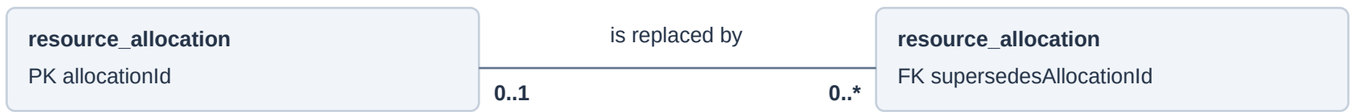
resource_allocation | Member C

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**allocationId**	**int**
FK	deploymentId	int
FK	reservationItemId	int
FK	resourceId	int
	capacityFraction	decimal
	startsAt	timestamp
	endsAt	timestamp (optional)
	allocationStatus	text
FK	supersedesAllocationId	int (optional)
	changeReason	text
FK	assignedByUserId	int (optional)

Relationships (parent to child)



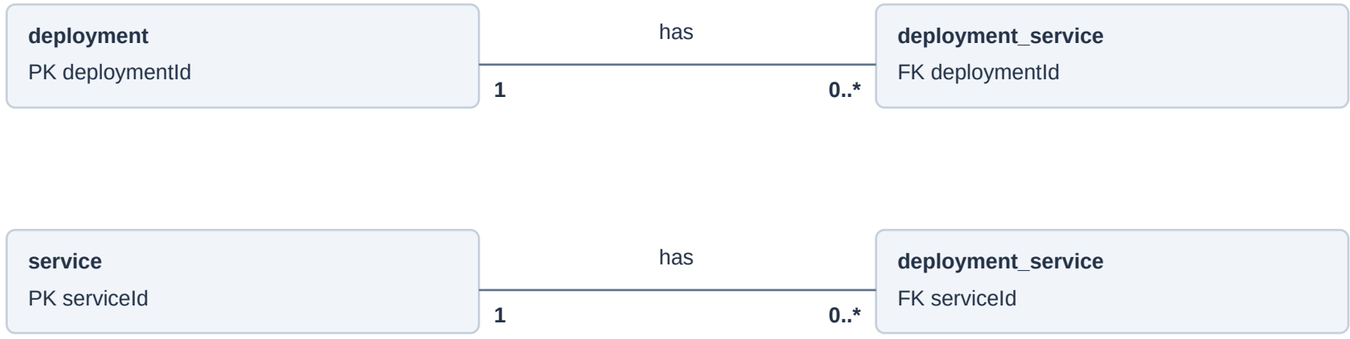


deployment_service | Member C

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**deploymentServiceId**	**int**
FK	deploymentId	int
FK	serviceId	int
	startsAt	timestamp
	endsAt	timestamp (optional)

Relationships (parent to child)



service | Member C

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**serviceId**	**int**
	name	text
	serviceType	text
	billingUnit	text

Relationships (parent to child)

No outgoing foreign keys; incoming links appear on child pages.

service_rate | Member C

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**serviceRateId**	**int**
FK	serviceId	int
	validFrom	timestamp
	validTo	timestamp (optional)
	unitPrice	decimal

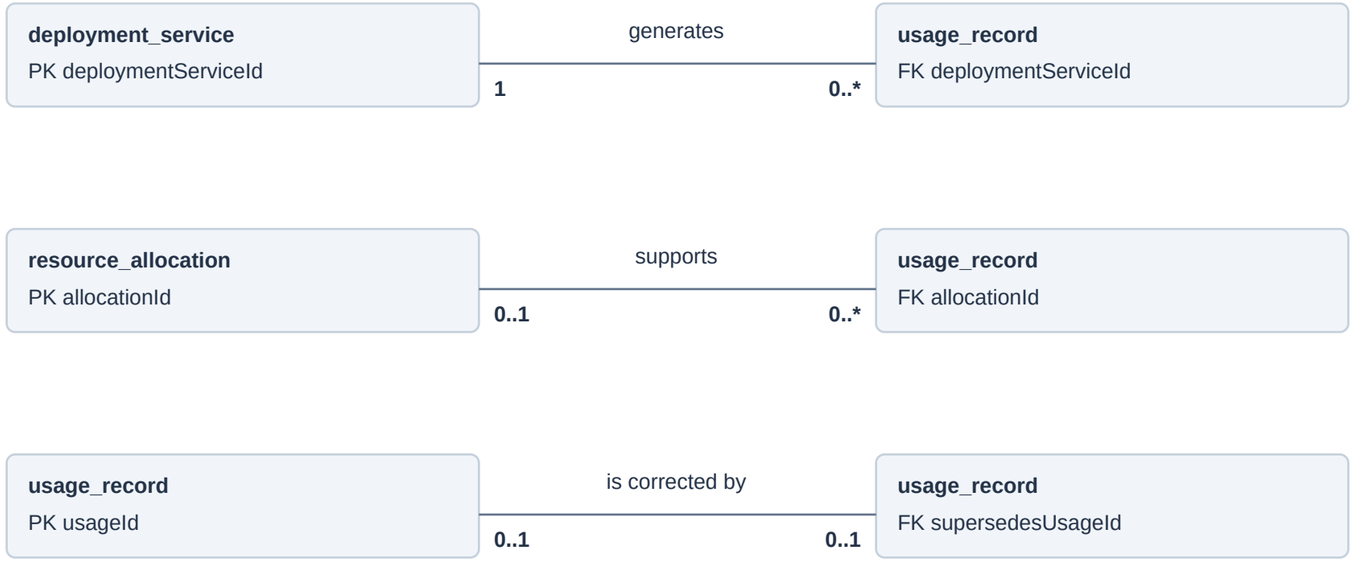
Relationships (parent to child)

usage_record | Member D

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**usageId**	**int**
FK	deploymentServiceId	int
FK	allocationId	int (optional)
	quantity	decimal
	periodStart	timestamp
	periodEnd	timestamp
	recordType	text
	recordedAt	timestamp
FK	supersedesUsageId	int (optional)

Relationships (parent to child)

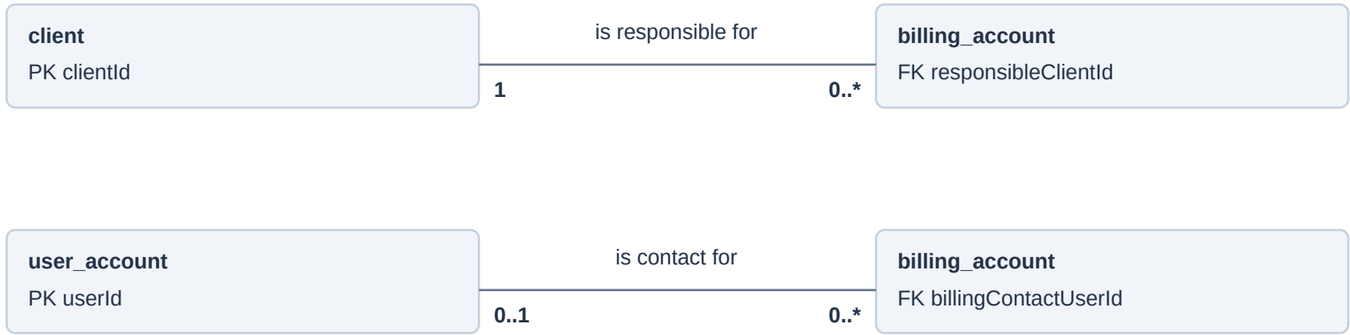


billing_account | Member D

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**billingAccountId**	**int**
FK	responsibleClientId	int
	name	text
FK	billingContactUserId	int (optional)

Relationships (parent to child)

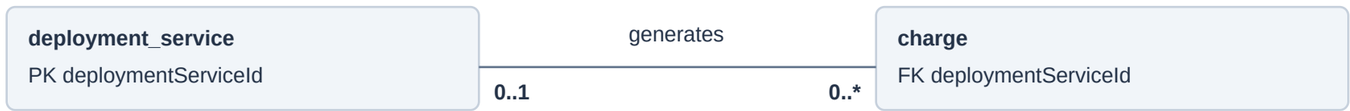


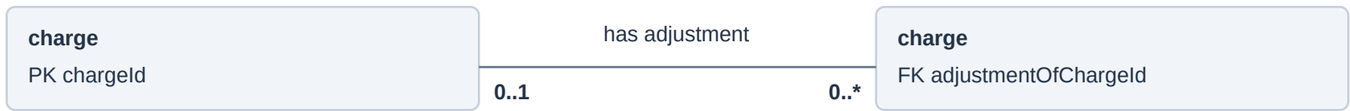
charge | Member D

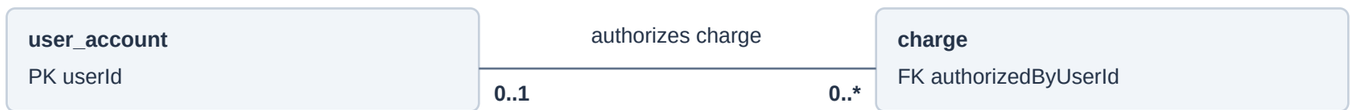
Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**chargeId**	**int**
FK	usageId	int (optional)
FK	deploymentServiceId	int (optional)
FK	serviceRateId	int (optional)
FK	adjustmentOfChargeId	int (optional)
FK	authorizedByUserId	int (optional)
	chargeType	text
	amount	decimal
	status	text
	reason	text
	createdAt	timestamp

Relationships (parent to child)







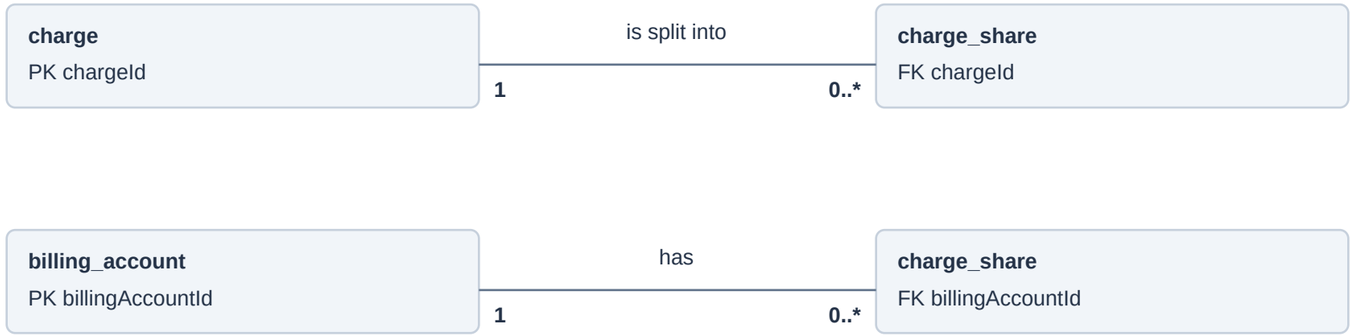
Group 4 | 1 = exactly one | 0..1 = optional one | 0..* = zero or more | Assumptions: analysis R1-R9

charge_share | Member D

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**chargeShareId**	**int**
FK	chargeId	int
FK	billingAccountId	int
	allocatedAmount	decimal

Relationships (parent to child)



invoice | Member D

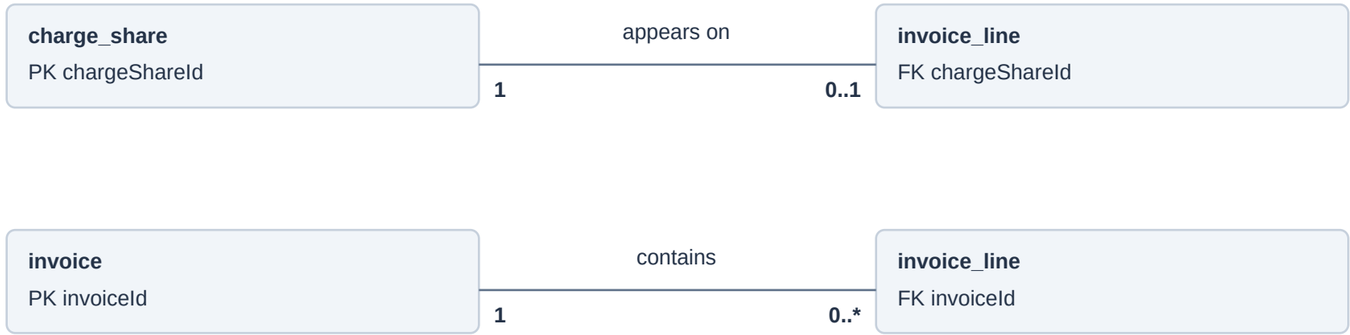
Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK	**invoiceId**	**int**
FK	billingAccountId	int
	issuedAt	timestamp
	dueAt	timestamp
	status	text

Relationships (parent to child)

invoice_line | Member D

Attributes and labelled relationships | One independent lane per foreign key

Key	Attribute	Type / optionality
PK/FK	**chargeShareId**	**int**
FK	invoiceId	int

Relationships (parent to child)